Oluwasola Oyinloye

Geospatial data scientist · Geospatial AI specialist

GIS · Remote sensing · Drone mapping · Geospatial AI

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I'm a geospatial data scientist and geospatial AI specialist in Lagos with five years of professional GIS and remote sensing work. I take on drone surveys in Nigeria, planned and flown in-house, and work remotely from satellite data for clients across Africa and beyond. Every result I deliver comes with how it was validated and the code that produced it.

SERVICES

- Geospatial AI and machine learning
- Drone mapping and photogrammetry
- Satellite remote sensing
- EUDR geolocation and deforestation checks
- GIS mapping and spatial analysis
- Flood, climate and environmental risk
- Web GIS development and dashboards
- Spatial databases and data engineering
- Training and capacity building

SELECTED WORK

Oil palm counting from aerial imagery

For a commercial oil palm plantation: a YOLOv8 crown-detection pipeline that mapped 718,751 palms across 12,166.6 ha in 26 blocks. Overlapping deliveries were resolved so every hectare is counted once, removing 302,412 double counts.

GeoAID flood susceptibility, Amuwo-Odofin (Lagos)

An open model with a live dashboard: structural flood susceptibility, near-real-time rainfall and exposure of people, schools, clinics and roads. ROC-AUC about 0.798 on a random split and about 0.642 under spatial cross-validation, reported side by side.

Lagos Access Map (in development)

A ward-by-ward audit of how complete OpenStreetMap's record of health facilities is in Lagos, with confidence intervals, as the base for access-to-care mapping.

SECTORS

- Agriculture and plantations
- Agric lending and finance
- Real estate, construction and infrastructure
- Government and urban planning
- NGOs, health and development
- Environment, climate and insurance
- Mining, quarrying and energy

WHERE

On site in Nigeria. Remote across Africa and worldwide.

WAYS TO WORK

Pilot · Project · Monitoring · Advisory days · Training. Written scope for every engagement; NDA on request.

TOOLS

Programming and analysis: Python, SQL, Jupyter, pandas, NumPy, SciPy, statsmodels. **Machine learning and AI:** PyTorch, Ultralytics YOLOv8, scikit-learn, XGBoost, SHAP, OpenCV. **Earth observation:** Google Earth Engine, geemap, Sentinel-1 SAR, Sentinel-2, SRTM, CHIRPS, NASA GPM IMERG, ESA WorldCover, WorldPop. **GIS and spatial data:** QGIS, Google Earth Pro, PostGIS, GeoPandas, Rasterio, Shapely, PyProj, OpenStreetMap. **Drone mapping:** DJI Terra, Agisoft Metashape, WebODM / OpenDroneMap, GNSS ground control. **Dashboards and web maps:** Streamlit, Power BI, Plotly, Folium, Leaflet.

EDUCATION

- Master of Information Technology (MIT), AI specialisation, Miva Open University
- BSc Surveying and Geoinformatics, University of Ilorin

TEACHING AND RESEARCH

Taught Python for geospatial analysis to 1,500 people through seminars.

Preprint: Oyinloye, O. A. and Mustapha, I. O. (2026). GeoAID: An Explainable Geospatial Disaster Intelligence Framework for Urban Flood Risk in a Data-Poor Coastal Megacity. Preprints.org. doi.org/10.20944/preprints202609.1630.v1